

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th March 2018

Subject ST2 – Life Insurance

Time allowed: Three hours (10.15* – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) A leading life insurance company has been so far selling only unit linked and term assurance contracts. The company now wishes to launch regular premium with-profit endowment assurance contracts.

- i)** State the needs of the customers that may be met by the regular premium with-profit endowment. (4)

During a Product Committee meeting, it was highlighted that once the new with-profit endowment assurance product is launched it is intended to discontinue the sale of existing unit linked contracts and to sell only the new with-profit endowment and existing term assurance contracts in the future.

- ii)** Outline the additional risks to which the Company will be exposed as a result of this change in product strategy. (8)

It has been suggested that the proposed with-profit endowment contract includes a guaranteed addition of 2% of sum assured each year in lieu of reversionary bonus with the balance to be paid in the form of terminal bonus

- iii)** Discuss this proposal from the points of view of both the insurer and the policyholder. (11)

The Chief Actuary of the Company has proposed a cost of guarantee charge to the with-profit fund to help the Company meet the cost of the guarantee being offered. The Chief Actuary has asked you to help in deciding the extent of cost of guarantee charge that can be levied to the asset share.

- iv)** State the type of model that you will choose, giving a reason for your choice (2)

- v)** Discuss the steps that are likely be involved in deciding the cost of guarantee charge. (7)

- vi)** State the cash flows, assumptions and parameters that you will consider in deciding the cost of guarantee charge. (4)

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Q. 2) A life insurance company is in the process of reviewing its experience analysis of assumptions from the past five years.

- i)** List the factors by which the persistency experience could be analyzed. (4)

- ii)** Describe the other considerations that would need to be taken into account before deciding which factors to use in the persistency experience analysis. (2)

In the last five years there has been a large capital investment by the company in developing a new administration system.

- iii)** Describe how the expenses relating to the introduction of this new administration system could be considered in analysis of the expense experience. (4)

The results of the mortality experience investigation for one product show that the pricing assumptions for mortality are significantly different from the past five years' experience.

- iv)** Suggest the next steps that the company could take given this observation. (7)

[17]

- Q. 3) i)** List the various components of asset share (2)

A life insurance company is now planning to calculate the asset share for its existing with-profit fund.

- ii)** Outline the checks the company will need to perform before reporting the numbers. (8)

It has been observed that the value of free assets in its with-profit fund is reducing each year for its existing business.

- iii)** Suggest reasons why the value of free assets could be falling each year. (8)
[18]

- Q. 4)** A life insurance company sells an existing term assurance product with age-specific premium rates and underwriting requirements.

The government has announced the roll-out of a “Universal Term Plan” which will be a term assurance plan available to all the citizens up to the age of 60. All the life insurance companies currently selling term assurance business in the country will be required to offer the Universal Term Plan. The sum assured and premium under the plan are fixed and will be the same for all ages. There will be no underwriting permitted under the product.

- i)** Discuss the possible impact on the existing term assurance portfolio of the company due to the introduction of the Universal Term Plan by the central government. (8)

The company is now considering suitable reinsurance strategies for the Universal Term Plan, and is looking at using stop loss or proportional reinsurance arrangements.

- ii)** Compare the suitability of stop loss and proportional reinsurance arrangements for the Universal Term Plan. (7)

- iii)** Discuss the issues that should be considered, before deciding on the reinsurance structure for the universal term plan. (8)
[23]

- Q. 5)** A company has been experiencing poor persistency at certain durations of policy term. The company is now considering alternatives that can be offered, where possible, to those lapsing the policies, so that the business can be retained by offering alternate terms.

- i)** State the possible alternatives that could be offered to policyholders who wish to lapse or surrender. (2)

- ii)** Give two examples for each of the following, of possible options that may be offered under a policy that could:
a) Increase mortality anti-selection; (2)
b) Increase financial anti-selection. (2)
[6]
